L–S–626c
May 2, 1967

SUPERSEDING
Int. Fed. Spec. L–S–00626b (COM–NBS)
June 19, 1961 and
Fed. Spec. L–S–626
January 7, 1943

FEDERAL SPECIFICATION

SPONGES, SYNTHETIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers sponges made of regenerated cellulose or synthetic resins.

1.2 Classification. Sponges furnished under this specification shall be of the following types, porosities, classes, and sizes, as specified (see 6.2):

Type I —Synthetic plastic.
Type II—Regenerated cellulose.
Porosity A—Coarse pores.
Porosity B—Fine pores.
Class 1—Uncompressed.
Class 2—Compressed.

1.2.1 Sizes. Sponges covered by this specification shall be of the dimensions or sizes shown in table II, as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

O–A–51 —Acetone, Technical.
O–H–765 —Hydrochloric Acid (Muriatic), Technical.
O–S–642 —Sodium Phosphate, Tribasic, Anhydrous; Dodecahydrate; and Monohydrate; Technical.
O–S–809 —Sulfuric Acid, Technical.
O–S–598 —Sodium Hydroxide, Technical.
TT–T–291 —Thinner; Paint, Volatile Spirits (Petroleum-Spirits).
VV–B–231 —Benzene (Benzol), Technical.
VV–G–101 —Gasoline; Motor, U. S. Government.
CCC–T–191 —Textile Test Methods.
PPP–B–636 —Box, Fiberboard.

Federal Standards:

Fed. Std. No. 123—Marking for Domestic Shipment.

FSC 7920

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials

3.1.1 Type I, plastic sponges. The plastic sponge shall be made of a polymerized synthetic resin, and such compounding ingredients as necessary to produce a soft pliable sponge when wet, conforming to the requirements hereinafter specified.

3.1.2 Type II, cellulose sponges. The cellulose sponge shall be made of a regenerated cellulose, and such compounding ingredients as necessary to produce a soft, pliable sponge when wet, conforming to the requirements hereinafter specified.

3.2 Chemical and physical properties. Sponges shall meet the requirements in table I when tested as specified in 4.3.6. Both types shall be furnished dry in accordance with industry practice.

Table I. Chemical and physical properties

Property	Type I	Type II
Water absorption, Minimum, percent by weight		
Porosity A	1500	1500
Porosity B	2000	1750
Ash, percent by weight (max)	5.0	5.0
pH range	6.0—9.0	6.0—9.0
Tensile strength, minimum psi, original specimen		
Length	10	25
Width	10	25
Solubility, percent by weight, maximum		
Gasoline	0.5	—
Benzene	1.0	—
Acetone	2.0	—
Mineral spirits (paint thinner)	1.0	—

3.2.1 Water resistance. The sponge shall show no change in texture, softness or volume from the original sponge when wet, when tested as specified in 4.4.5 and 4.4.6.

3.2.2 Resistance to solvents (type I only). The type I sponge shall not be soluble in the solvents in table I to an extent greater than indicated, when tested as specified in 4.4.7.

3.2.3 Resistance to acids and alkalies. (Applicable to type I only.) The type I sponge shall show no deterioration of the sponge material or change in texture or softness, nor shall the tensile strength of the

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test specimens decrease more than 10 percent from that of the original specimens, when subjected to solutions of hydrochloric acid, sulfuric acid, sodium hydroxide to trisodium phosphate as described in 4.4.8 and 4.4.9.

3.2.4 Color. Unless otherwise indicated in the invitation for bids, sponges shall be furnished in buff (natural) color.

3.3 Porosity. Sponges furnished shall be coarse pored (porosity A) or fine pored (porosity B), as specified (see 6.2), and shall have the water absorption indicated in table I. Porosity A sponges shall have pores ranging from 3/32 to 3/4 inch in diameter, and porosity B sponges shall have pores 3/32-inch or less in diameter.

3.4 Sizes. Measurements for size shall be made on the moist sponge. All sponges, except type II size 5, shall be rectangular in shape and shall conform to the requirements in table II. Type II size 5 shall be cylindrical. A tolerance of ±1/4-inch in length and width and ±1/8-inch in thickness shall be permitted for the type I sponges. Type II sponges shall have the minimum volumes in table II, provided the dimensions do not fall below the minimum values indicated.

Table II. Dimensions and sizes of sponges

Type I—Synthetic plastic sponges

Thickness (inches)	Width (inches)	Length (inches)
1 1/8	3 3/4	5 3/4
1 1/2	4 3/8	7 3/8
1 5/8	4 3/8	5 7/8
1 7/8	3 1/2	4 3/4
2 1/4	4 3/8	7
2 1/2	4	6
2 5/8	6	8
2 7/8	7	9
2 3/4	4 3/4	6 1/2

Type II—Cellulose sponges

Size	Cubic inches (Min.)	Width (Min.)	Length (Min.)	Thickness (Min.)
1	13.8	2 3/4	4 3/8	7/8
2	22.5	3 1/2	4 5/8	1 1/4
3	43.5	3 5/8	5 3/4	1 3/4
4	68.0	4 1/4	6 1/4	2 1/8
5	3.5	2 1/4 (diameter)		1 3/8

3.5 Classes. Sponges shall be furnished in the uncompressed state (class 1) or in the compressed state (class 2), as specified (see 6.2). When class 2 sponges are specified, the thickness of the sponge after compression shall be no more than 20 percent of the thickness of the moist sponges indicated in table II.

3.6 Workmanship. Sponges shall be rectangular (except type II, size 5) and uniform in size and shape. They shall be cleanly cut, free of grit, dirt and extraneous matter and defects that affect appearance or impair serviceability.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Inspection responsibility. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other commercial laboratory acceptable to the Government. The Government reserves the right to perform any inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Sampling for inspection. Sampling for inspection shall be performed in accordance with MIL–STD–105, except as otherwise indicated. A lot for sampling purposes shall consist of all sponges of the same type, porosity, class and size offered for delivery at one time.

4.2 Inspection of components and materials. The supplier is responsible for insuring that materials and components used in the end item were manufactured, examined and tested in accordance with this specification or referenced subsidiary specifications to the extent specified.

4.3 Inspection of the end-item. Each lot of sponges shall be inspected for visual defects in accordance with 4.3.1; dimensional defects in accordance with 4.3.2; examined for count in accordance with 4.3.3; and for defects in preparation for delivery in accordance with 4.3.4. The lot size shall be expressed in units of sponges for examinations in 4.3.1 to 4.3.3 inclusive, and in units of shipping containers in 4.3.4. Inspection levels and acceptable quality levels (AQL's) in 4.3.5 shall apply.

4.3.1 Examination of the end-item for visual defects. Each sample unit shall be examined for visual defects in accordance with defects below. The sample unit shall be one sponge. Sponges shall be moistened until full growth is reached before examination.

Examine	Defect
Material	Not type, porosity or class specified.
	Not rectangular shaped, or cylindrical as required.
	Not natural buff color, or color specified.
	Color not uniform.
	Not soft and pliable when moist.
Workmanship	Frayed, cut or torn.
	Embedded grit; coarse particles or presence of loose, foreign or extraneous matter.
	Not clean.

4.3.2 Dimensional examination. The sample unit shall be one sponge. Each sample unit shall be examined for defects below.

Examine	Defect
Dimensions	Volume, length, width, diameter or thickness (as applicable) less than specified.
	NOTE: Sponges shall be immersed in water until absorption has effected fullest growth. Squeeze out excess water using normal hand-pressure and measure the dimensions.
Porosity	Using a scale graduated in $\frac{1}{64}$th inch, measure the diameters of 10 openings on each surface of the sponge and compare the average of the reading with the pore diameters specified.
Compression (class 2 only)	More than 20 percent of the thickness of the moist sponge.

4.3.3 Examination of the end-item for count. The sample unit for this examination shall be one (1) unit package or intermediate package, as applicable. The count shall be not less than the specified or indicated quantity.

4.3.4 Examination of the preparation for delivery. Each lot shall be examined to determine that the packaging, packing and marking comply with the requirements of section 5 of this specification. The lot size shall be the number of shipping containers in the lot. The sample unit for examination shall be one shipping container fully prepared for delivery.

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Examine	Defect
Packaging	Packaging material not as specified. Not unit packaged as specified. Not individually wrapped or in individual envelopes.
Packing	Not in accordance with contract requirements. Mixed type, porosity, class, shape, size or color in one shipping container. Number of unit packages per shipping container not as specified in contract or order. Container material not as specified; closures not accomplished by specified or required methods or materials. Container material broken, dented or otherwise damaged to the extent that serviceability will be affected.
Markings	Interior or exterior markings (as applicable) illegible, incorrect, incomplete, omitted or not in accordance with contract requirements.
Weight	Gross or net weight exceeds box requirements.

4.3.5 Inspection levels and Acceptable Quality Levels (AQL's) for examinations. The inspection levels for determining the sample sizes, and the acceptable quality levels (AQL's), expressed in defects per hundred units shall be as indicated below:

Inspection paragraph	Inspection level	AQL
4.3.1	S–4	4.0
4.3.2	S–3	4.0
4.3.3	S–2	4.0
4.3.4	S–2	4.0

4.3.6 Testing of the end-item. Each lot of sponges shall be tested for the characteristics in table III, as applicable. The sample unit for testing shall consist of ten (10) sponges selected at random from a shipping container. The sample size shall be three sample units each drawn from a different shipping container. There shall be no failure of any sample unit to meet the unit or lot average requirements for any characteristic.

4.4 Test methods

4.4.1 Water absorption. Wash 3 specimens of sponge, (2-inches by 2-inches by the thickness of the sponge) free of water soluble materials by filling with water and hand squeezing as dry as possible. Repeat the washing operation not less than five times, using distilled water for the last wash. Condition the washed specimens for 24-hours at 73°±2° F. and 50±4 percent relative humidity. Weigh the specimens to the nearest 10 milligrams and record the weight as W_1. Immerse the specimens in water maintained at room temperature, and squeeze the immersed specimens until all the air is expelled. Attach a fine wire hook to one corner of each specimen, withdraw from the water and suspend for 30 seconds to drain excess water. Place the specimens on separate tared watch glasses, remove the hooks and reweigh the wet specimens. Record the weight as W_2. Calculate the percent absorption as follows:

$$\text{Percent absorption} = \frac{W_2 - W_1}{W_1} \times 100$$

4.4.2 Ash. Wash approximately 2 grams of sponge free of soluble materials as described in 4.4.1 and dry to constant weight of 100°±2° C. Cool the specimen in a desiccator and weigt in a tared crucible. Record the weight as W_1. Place the crucible containing the dry specimen in a muffle furnace and gradually raise the temperature until the specimen chars but does not burn. After smoking ceases, raise the temperature until the carbonaceous material burns off and only ash remains. Remove the crucible, cool in a desiccator, reweigh and record the weight as W_2. Calculate the ash as follows:

$$\text{Percent ash} = \frac{W_2}{W_1} \times 100$$

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Table III. Instructions for testing

Characteristic	Specification reference		Requirement applicable to		Number determinations per sample unit	Results reported as	
	Requirement	Test method	Individual	Lot average		Pass or fail	Numerically to nearest
Chemical and physical properties (types I and II)							
Absorption	Table I	4.4.1	—	X	3	—	1.0 percent
Ash	Table I	4.4.2	—	X	3	—	0.1 percent
pH	Table I	4.4.3	—	X	1	—	0.1 pH
Tensile strength							
Length	Table I	4.4.4	—	X	3	—	0.1 pound
Width	Table I	4.4.4	—	X	3	—	0.1 pound
Solubility (type I only)							
(a) Gasoline	2.2.2	4.4.7	—	X	1	—	0.1 percent
(b) Benzene	2.2.2	4.4.7.1	—	X	1	—	0.1 percent
(c) Acetone	2.2.2	4.4.7.1	—	X	1	—	0.1 percent
(d) Mineral spirits	2.2.2	4.4.7.1	—	X	1	—	0.1 percent
Resistance to water							
Cold water	3.2.1	4.4.5	X	—	1	X	—
Hot water	3.2.1	4.4.6	X	—	1	X	—
Resistance to acids (type I only)							
(Hydrochloric and sulfuric)							
Material, softness and texture	3.2.3	4.4.8	X	—	1	X	—
Tensile strength	3.2.3	4.4.4	—	X	1	—	0.1 pound
Resistance to alkalies (type I only)							
(Sodium hydroxide and trisodium phosphate)							
Material, softness and texture	3.2.3	4.4.9	X	—	1	X	—
Tensile strength	3.2.3	4.4.4	—	X	1	—	0.1 pound

4.4.3 Acidity (pH). Immerse approximately 2 grams of sponge in 250 ml. of distilled water in a beaker and boil the solution for 10 minutes. Cool to room temperature and determine pH using a standardized pH meter.

4.4.4 Tensile strength. Immerse the sponge in water until it has expanded to full volume; remove from the water and squeeze out the excess water by hand. Cut specimens, 1-inch thick by 2-inches wide by 2-inches long, three in the width direction of the sponge and three in the length direction of the sponge. Determine the tensile strength of each specimen by method 5100 of CCC-T-191, except that the distance between the jaws shall be 1-inch. Calculate the tensile strength by dividing the breaking force in pounds by the cross sectional area of the specimen in square inches.

4.4.5 Resistance to cold water. Immerse a 1-inch cube of sponge in cold water (25±1° C.) for 24 hours. At the end of the immersion period remove the specimen; squeeze out the excess water by hand and set aside to dry in air. Rewet the specimen, squeeze dry by hand and compare with a damp original sponge for softness and texture.

4.4.6 Resistance to hot water. Immerse a 1-inch cube of sponge in 500 ml. of distilled water contained in an Erlenmeyer flask equipped with a reflux condenser. After 2 hours of refluxing, remove the specimen, squeeze out excess water by hand and set aside to air dry. Rewet the specimen, squeeze-dry as before and compare with a damp original sponge for texture and softness.

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4.4.7 Solubility in organic solvents. Wash approximately two grams of a sponge free of water soluble materials as in 4.4.1 and dry to constant weight in an oven maintained at 100°±2° C. Cool in a desiccator and weigh to the nearest milligram. Record the weight as W_1. Place the weighed specimen in a Soxhlet extraction apparatus and extract with gasoline conforming to VV-G-101, for not less than 3 hours, heating at a rate to fill the extraction tube not less than 12 times. Remove the specimen at the end of the extraction period and squeeze out the excess gasoline by hand. Dry the specimen to constant weight in the oven, cool, reweigh and record the weight as W_2. Calculate the percent solvent soluble material as follows:

$$\text{Percent solvent soluble material} = \frac{W_1 - W_2}{W_1} \times 100$$

4.4.7.1 Solubility in Benzene, Acetone and Mineral Spirits. Follow the procedure in 4.4.7 substituting the following solvents for the gasoline:

Benzene conforming to VV-B-231
Acetone conforming to O-A-51
Thinner conforming to TT-T-291

4.4.8 Resistance to hydrochloric and sulfuric acids. Place specimen of sponge, 2-inches by 2-inches by 1-inch in size, in different beakers and cover them with a 7-percent solution of hydrochloric acid, prepared from acid conforming to O-H-765, and a 20-percent solution of sulfuric acid prepared from acid conforming to O-S-809, respectively. After 24 hours immersion, remove the specimens, wash with water and examine for deterioration of the material or change in texture or softness as compared to an untreated specimen. Determine the tensile strength of the treated specimens in accordance with 4.4.4 and compare with the untreated sponge.

4.4.9 Resistance to alkalies. Follow the procedure 4.4.8 except that a 10-percent solution hydroxide, and 5-percent solution of trisodium phosphate prepared from reagents conforming to O-S-598 and O-S-642, respectively shall be used in place of the acids.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Sponges shall be individually wrapped or enveloped in cellophane, polyethylene or other similar transparent type material and sealed to prevent entry of water or foreign matter.

5.1.2 Level C. Sponges shall be packaged to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity. The supplier's commercial practice will be acceptable when it fulfills the above requirements.

5.2 Packing. Packing shall be levels A, B, or C, as specified (see 6.2).

5.2.1 Level A. Sponges packaged as specified in 5.1 and in the quantities specified (see 6.2), shall be packed in fiberboard boxes conforming to PPP-B-636, style RSC, type CF or SF, V3c or V3s. Caseliners will not be required, however, boxes shall be waterproofed and closed in accordance with the appendix to the box specification. The gross weight of the shipping container shall not exceed the limitations of the box specification.

5.2.2 Level B. Sponges packaged as specified in 5.1 and in the quantities specified (see 6.2), shall be packed in fiberboard boxes conforming to PPP-B-636, style RSC, type CF or SF, class domestic, variety SW, grade 275. Box closure and gross weight requirements shall be in accordance with the appendix to the box specification.

5.2.3 Level C. Sponges, packages as specified in 5.1 shall be packed to insure carrier acceptance and safe delivery at destination at the lowest transportation rates for such supplies. Containers shall be in accordance with rules and regulations of carriers applicable to the mode of transportation.

5.3 Marking.

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5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military requirements. In addition to any special markings required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 *Intended use.* Type I sponges are intended for general cleaning operations necessitating use of acid or alkaline solutions or organic solvents. Type II sponges are intended for general cleaning operations using commercial detergents, soaps, and cleaning agents. Type II porosity B fine pored sponges are intended for photographic applications. Type II, circular sponges are intended for use in sponge cups and by binders and counters.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, porosity, and class required (see 1.2).
(c) Size required (see 3.4).
(d) When a particular color is required (see 3.2.4).
(e) Selection of the applicable level of packaging and packing (see 5.1 and 5.2).
(f) Number of sponges per shipping container (see 5.2.1 and 5.2.2).

6.3 Items procured under this specification for military use are to be limited to the variety shown on the applicable military standards. Personnel of the military departments are requested to refer to these documents for guidance.

CUSTODIANS:

Army—GL
Navy—SA
Air Force—WRAMA (84)

Preparing activity:

Navy—SA

Civilian agencies interest:

Review activities:

Army—GL
Navy—SA
Air Force—WRAMA (84)

User activities:

Army—
Navy—
Air Force—

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 5 cents each.

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